

Statistical Methods 1: Bayes Theorem

Why study probability and statistics?

- If an eclipse was once modeled as an event with a certain probability each day, and now we can predict them, then the model has changed from **stochastic** to **deterministic**.
- Probability gives us a way to model uncertainty, and statistical inference allows us to evaluate our confidence in a particular model.

Mathematical Formalism

With any experiment, we have an associated **sample space** Ω consisting of **outcomes** $\{O_1, O_2, \dots, O_m\}$.

- E.g. A coin toss can be either heads or tails: $\Omega = \{H, T\}$

Each outcome is assigned a probability, according to the physical understanding of the experiment.

A **Discrete Sample Space** is a sample space with a countable infinite or finite number of elements O_i , which can be given a probability P_i such that:

$$\sum_{i=1}^m P_i = 1$$

Events are subspaces of Ω , which are assigned a probability.

A function which maps $\Omega \rightarrow \mathbb{R}$ is called a **Random Variable**.

Example

- Let X be the amount of money won on a die throw

Outcome (w)	1	2	3	4	5	6
Net dollars earned	-8	2	0	4	-2	4

- We are interested in events of the form:

$$\{\omega \in \Omega \mid X(\omega) = x\}, \quad \text{for a given } x \quad (\text{shorthand: } \{X = x\})$$

- For $x = 0$: $\{X = 0\}$ is the event: $\{3\}$
- For $x = 4$: $\{X = 4\}$ are the events: $\{4, 6\}$
- For $x = 60$: there are no events ($\{X = 60\} = \emptyset$)
- Here X is the **random variable** and x is some value attainable by X .

The probabilistic properties of a random variable are determined by the probabilities assigned to the outcomes of the underlying sample space.

Example

- $P(\{X = 4\})$ is the probability of the event $\{X = 4\}$

- $P(\{\omega \in \Omega \mid X(\omega) = 4\}) = P(\{4, 6\}) = P_4 + P_6 = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$
- In statistical mechanics, an event $\{X = 4\}$ would describe a macrostate with two ($\{4, 6\}$) microstates.

Inclusion-Exclusion rule: For any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Axiom — Finite Additivity: Given disjoint events A and B :

$$P(A \cup B) = P(A) + P(B)$$

You can't roll a 4 and a 6 when rolling one six sided die, but the probability of it being a Monday or before noon is $\frac{1}{2} + \frac{1}{7} - \frac{1}{2 \times 7}$

Probability Mass Function

$f_X(x)$ can be obtained by summarizing the nonzero values of probabilities assigned to events ($P(X = x)$).

Properties:

- Probabilities are positive semi-definite: $P(A_i) \geq 0$
- Exactly one event should happen: $P(\Omega) = 1$
- $P(A_i \cup A_j) = P(A_i) + P(A_j)$: Since a die cannot land on 4 and 6, the probability of rolling a 4 OR a 6 is the probability of rolling a 4 plus the probability of rolling a 6.

Example

- The probability of earning some amount of money on our die roll from above can be described as:

$$f_X(x) = P(\{\omega \in \Omega \mid X(\omega) = x\}) = \begin{cases} 1/6 & \text{for } x = 0 \\ 1/6 & \text{for } x = 2 \\ 2/6 & \text{for } x = 4 \\ 1/6 & \text{for } x = -2 \\ 1/6 & \text{for } x = -4 \\ 0 & \text{for any other value} \end{cases}$$

Conditional Probability

Given the information that event B has occurred, the probability that A occurs is called the **conditional probability** of A given B , or $P(A \mid B)$:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \iff P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$$

This is called the **Multiplicative law of conditional probability**

The Law of total probability states:

$$P(A) = \sum_i P(A | B_i)P(B_i)$$

where B_i is a "partition" of the sample space such that $\sum P(B_i) = 1$.

Bayes' Theorem

Suppose A occurs. Let $\{B_i\}$ be all possible scenarios where A occurs. If we know $P(B_i)$, our prior, we can find $P(B_i | A)$, the probability that B_i happened along with A , or that A happened because of B_i :

$$P(B_i | A) = \frac{P(A \cap B_i)}{P(A)} \quad \# \text{ multiplicative law of conditional probability}$$

$$\boxed{P(B_i | A) = \frac{P(A | B_i)P(B_i)}{P(A)}} \quad \# \text{ This is Bayes' theorem}$$

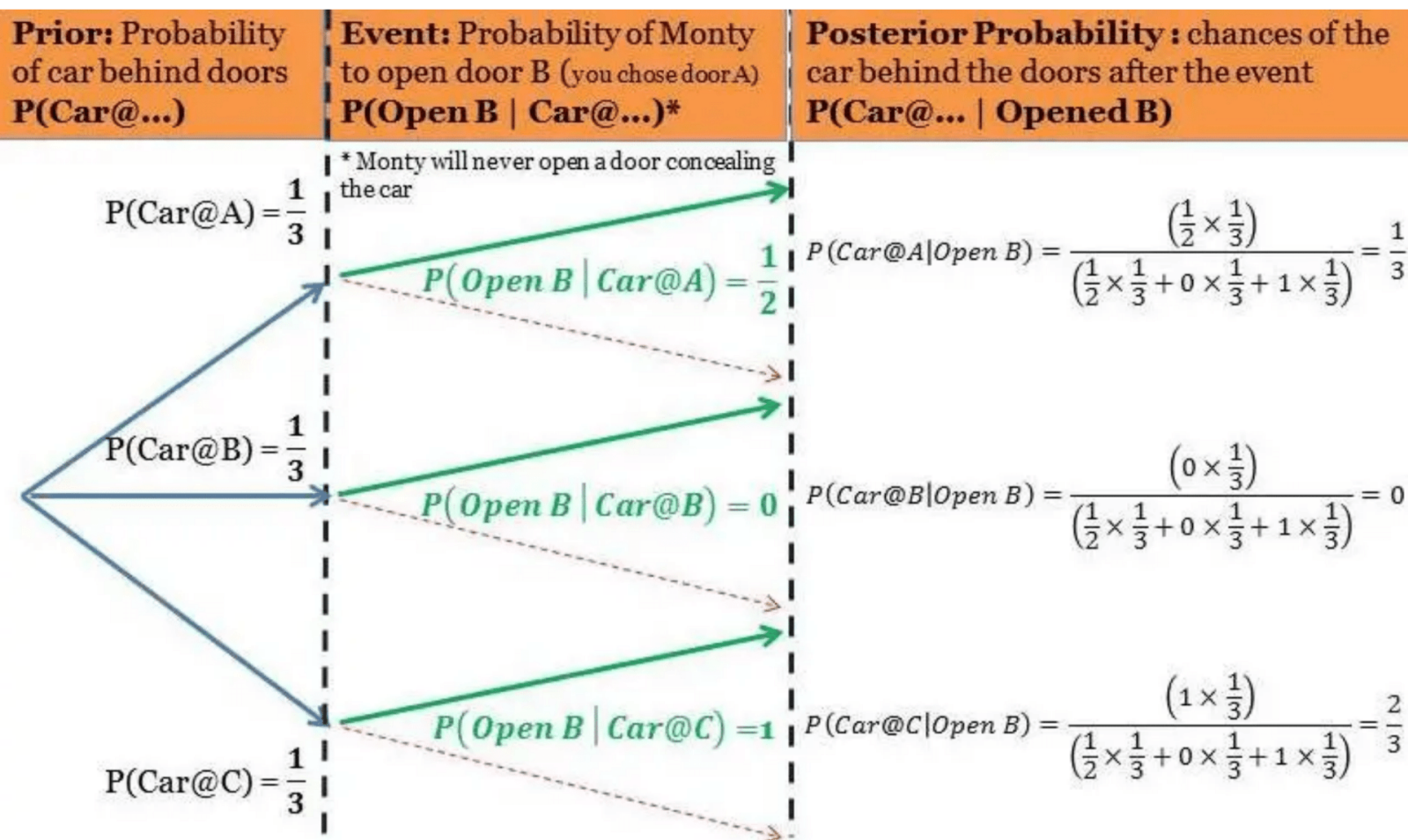
- $P(A | B_i)$ is called a likelihood
- $P(B_i)$ is called a prior
- $P(B_i | A)$ is called a posterior probability

Another form of Bayes' Theorem:

$$P(B_i | A) = \frac{P(A | B_i)P(B_i)}{\sum_{j=1}^m P(A | B_j)P(B_j)}$$

Activity - The Monty Hall Problem

Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, 'Do you want to pick door No. 2?' Is it to your advantage to switch your choice?



Resources

- [PennState Center for Astrostatistics and Astroinformatics](#) (online resources no longer available inspired this lecture, but they have other online resources)
- [Bayesian Methods for Hackers](#): The book I would teach a full course out of if given the opportunity. Has Jupyter notebooks and resources online that accompany the textbook.
- [Barford: Experimental Measurements: Precision, Error, and Truth](#) is all the frequentist statistics you will need to teach experiments in modern physics.
- A good general machine learning textbook for statistical and computational methods: [Hastie, Tibshirani, and Friedman: The Elements of Statistical Learning: Data Mining, Inference and Prediction](#)
- [Cornell Page on the Monte Hall Problem](#)
- [3Blue1Brown](#) is an excellent YouTube channel to learn about statistics and machine learning. I highly recommend his videos on [The Central Limit Theorem](#), and [Solving Wordle using information theory](#).
- [Maya's Notebook on Data Analysis](#)
- [Maya's Notebook on Metropolis-Hastings](#)